Project Proposal

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Team Information

Team members: Max Yin, Jenni Tong, Yingming Sha

Public repository:
https://github.com/ymyms/stat-405-project

The repository contains commits from all team members and will be used to store data, code and the report.

Candidate Datasets

Main Dataset: NBA Players Dataset

URL:
https://www.kaggle.com/datasets/justinas/nba-players-data

This dataset contains information on players who have appeared on NBA team rosters over more than two decades. Each observation corresponds to a player-season entry and includes demographic variables such as age, height, weight, and place of birth, along with biographical details like the team played for, draft year, and draft round. The dataset also includes basic box score statistics such as games played and average numbers of points, rebounds, and assists. Together, these variables provide both physical characteristics of players and measures of their performance.

```r
nba <- read.csv("all_seasons.csv")
head(nba)
```

```
##   X      player_name team_abbreviation age player_height player_weight
## 1 0 Randy Livingston               HOU  22        193.04      94.80073
## 2 1 Gaylon Nickerson               WAS  28        190.50      86.18248
## 3 2     George Lynch               VAN  26        203.20     103.41898
## 4 3   George McCloud               LAL  30        203.20     102.05820
## 5 4     George Zidek               DEN  23        213.36     119.74829
## 6 5   Gerald Wilkins               ORL  33        198.12     102.05820
##                 college country draft_year draft_round draft_number gp  pts reb
## 1       Louisiana State     USA       1996           2           42 64  3.9 1.5
## 2 Northwestern Oklahoma     USA       1994           2           34  4  3.8 1.3
## 3        North Carolina     USA       1993           1           12 41  8.3 6.4
## 4         Florida State     USA       1989           1            7 64 10.2 2.8
## 5                  UCLA     USA       1995           1           22 52  2.8 1.7
## 6 Tennessee-Chattanooga     USA       1985           2           47 80 10.6 2.2
##   ast net_rating oreb_pct dreb_pct usg_pct ts_pct ast_pct  season
## 1 2.4        0.3    0.042    0.071   0.169  0.487   0.248 1996-97
## 2 0.3        8.9    0.030    0.111   0.174  0.497   0.043 1996-97
## 3 1.9       -8.2    0.106    0.185   0.175  0.512   0.125 1996-97
## 4 1.7       -2.7    0.027    0.111   0.206  0.527   0.125 1996-97
## 5 0.3      -14.1    0.102    0.169   0.195  0.500   0.064 1996-97
```

```
## 6 2.2       -5.8    0.031    0.064   0.203  0.503   0.143 1996-97
```

This dataset is suitable for our project because it includes both player attributes and performance statistics that can be analyzed using Bayesian. In particular, variables such as height and weight allow us to study the relationship between players' physical characteristics. Since the dataset contains many observations across multiple seasons, it provides enough data to estimate posterior distributions and compare different Bayesian modeling approaches.

Backup Dataset: Olympic Athletes Dataset

URL:
https://www.kaggle.com/datasets/abdullahmeo/120-years-of-olympic-athletes-dataset

This dataset contains information on athletes who participated in the modern Olympic Games from Athens 1896 to Rio 2016, covering more than 120 years of Olympic history. The dataset includes 271,116 observations and 15 variables, where each row corresponds to an athlete competing in a specific Olympic event. The variables include demographic characteristics such as sex, age, height, and weight, as well as information about the athlete's team, country (NOC code), sport, and event. The dataset also records the year and location of the Olympic Games, along with medal outcomes (Gold, Silver, Bronze, or no medal). Overall, the dataset provides a broad overview of athlete characteristics and participation across many sports and countries over time.

```r
olympic <- read.csv("athlete_events.csv")
head(olympic)
```

```
##   ID                     Name Sex Age Height Weight           Team NOC
## 1  1                A Dijiang   M  24    180     80          China CHN
## 2  2                 A Lamusi   M  23    170     60          China CHN
## 3  3      Gunnar Nielsen Aaby   M  24     NA     NA        Denmark DEN
## 4  4     Edgar Lindenau Aabye   M  34     NA     NA Denmark/Sweden DEN
## 5  5 Christine Jacoba Aaftink   F  21    185     82    Netherlands NED
## 6  5 Christine Jacoba Aaftink   F  21    185     82    Netherlands NED
##         Games Year Season      City         Sport
## 1 1992 Summer 1992 Summer Barcelona    Basketball
## 2 2012 Summer 2012 Summer    London          Judo
## 3 1920 Summer 1920 Summer Antwerpen      Football
## 4 1900 Summer 1900 Summer     Paris    Tug-Of-War
## 5 1988 Winter 1988 Winter   Calgary Speed Skating
## 6 1988 Winter 1988 Winter   Calgary Speed Skating
##                                Event Medal
## 1        Basketball Men's Basketball  <NA>
## 2       Judo Men's Extra-Lightweight  <NA>
## 3            Football Men's Football  <NA>
## 4        Tug-Of-War Men's Tug-Of-War  Gold
## 5   Speed Skating Women's 500 metres  <NA>
## 6 Speed Skating Women's 1,000 metres  <NA>
```

This dataset serves as a suitable backup for our project because it also contains physical attributes such as height and weight, along with measurable outcomes such as medal results. These variables allow similar Bayesian analyses to those proposed for the NBA dataset, such as modeling relationships between physical characteristics and athletic performance. In addition, the large number of observations across many years provides sufficient data for estimating posterior distributions and comparing Bayesian models if we choose to use this dataset.

Scientific Questions (will be decided after further discussion)

Our main goal is to study relationships between physical attributes and basketball outcomes using Bayesian inference.

Our primary scientific question is:

```
recent_10 <- nba[nba$draft_year>=2006 & !(nba$draft_round== "Undrafted"),]

rookie_data <- recent_10 |>
  dplyr::arrange(player_name, season) |>
  dplyr::distinct(player_name, .keep_all = TRUE)

quantile(rookie_data$player_height,c(0.25,0.5,0.75))
```

```
##    25%    50%    75%
## 195.58 200.66 208.28
```

What is the posterior distribution of the slope parameter describing the relationship between NBA player height and weight?

Equivalently, we want to estimate the posterior distribution of the slope coefficient in a regression model of player weight on height, as well as the posterior expectation $E[\beta \mid \text{data}]$.

A secondary question we may investigate is:

How does player height affect the probability that a player averages more than 20 points per game?

These questions are well suited to Bayesian methods because they allow us to quantify uncertainty, incorporate prior information, and compare posterior distributions under different model assumptions.

Proposed Methodology

Model 1:

- Perform Exploratory Data Analysis to visualize, clean ,and explore potential relationship between variables.
- Assign prior and likelihood distribution based on observations, visualizations, or knowledge from prior studies.

Posterior Computation Plan

We plan to use at least two methods for posterior computation:

1. SNIS
2. MCMC

This will allow us to compare posterior approximations and assess their agreement. We will discuss if the results support one another, and why such agreement or disagreement happens.

Model Checking and Validation

We plan to evaluate our Bayesian models using:

- Knowledege from prior research studies
- Check if the credible interval is at least approximatedly calibrated

- Visualize the posterior distribution

Team Contribution Plan

To ensure all team members contribute roughly equally, we plan the following division of work:

- **Max Yin:**

  Exploratory Data Analysis

  Discussion and Limitations

  Compare the results from the two methods

- **Jenni Tong:**

  Constructing Prior and likelihood Distribution

  Implement MCMC method

  Analyze posterior distribution

- **Yingming Sha:**

  Constructing Posterior Distribution

  Implement SNIS method

  Interpret posterior result

All team members will contribute to background intriduction, literature review, reference